

scripts/run-challenge-matrix.sh

— User Guide

Last verified: 2026-07-26 (LVA-014 durable device-gate fixes — AVD-name resolution, WaitForBoot container-liveness, emulator-image preflight) **Inheritance:** HelixConstitution §11.4.18 + Lava §6.AE.6 + §6.X (Container-Submodule Emulator Wiring) + §6.I (Multi-Emulator Container Matrix)

Overview

Operator entry point for §6.AE gate-mode Challenge Test runs. Pre-bakes the §6.AE.2 minimum AVD matrix (API 28 / 30 / 34 / latest stable × phone + tablet) and delegates to submodules/
containers/cmd/emulator-matrix --runner=containerized per §6.X.

LVA-014 durable device-gate fixes (2026-07-26)

The 2026-07-04 "device gate boot hang" was diagnosed as an **AVD-name mismatch**: the baked emulator images (ghcr.io/vasic-digital/lava-android-emulator:apiNN-x86_64) bake exactly ONE AVD named default, but the runner passed the §6.AE.2 matrix names (CZ_API34_Phone, ...) verbatim. The image entrypoint exited in ~4s ("AVD not found"), the container's --rm reaped the log, and WaitForBoot polled the dead forwarded port to its deadline, misreporting the 4s exit as a boot timeout. Three durable fixes:

1. **AVD-name resolution** (submodules/containers/pkg/emulator/avdresolve.go, wired into Containerized.Boot) — the runner queries the AVDs actually baked into each image (avdmanager list avd inside a throwaway --rm --entrypoint avdmanager container) and maps the requested name:api:form entry to a real baked AVD. Exact name match → used verbatim (the proven --avds "default:34:phone" workaround is byte-compatible). Api-level match → the baked name is substituted with a stderr note (the requested name was advisory). No match → Boot fails FAST, before any container is launched, naming the available baked AVDs.

2. **Container liveness in WaitForBoot** (submodules/containers/pkg/emulator/containerized.go) — every poll iteration first checks podman/docker inspect that the emulator container is still running; on exit it fails immediately with the container's captured logs (<runtime> logs --tail 100, fetched before --rm reaps them; an already-reaped container is reported honestly). EMU-1 semantics are preserved: a Containerized without a Boot-set container name performs no liveness exec, and every liveness exec is bound to the WaitForBoot deadline context.
3. **Emulator-image preflight** (this script, containerized runner only) — before building the APK, the script instantiates the {api} image template per distinct api level in the matrix, checks each image with <runtime> image inspect, and pulls the missing ones with live progress. A pull failure is an honest operator-facing error (exit 1) that names the image, shows the pull error tail, and prints the local-build fallback command from submodules/containers/pkg/emulator/Containerfile — never a silent skip.

The default --container-image is now the template ghcr.io/vasic-digital/lava-android-emulator:api{api}-x86_64; the runner substitutes {api} with each AVD's api level at Boot time, so one reference covers the whole multi-api matrix. A --container-image without {api} is used verbatim for every AVD (the pre-LVA-014 single-image behavior).

Honest pre-flight

The script DETECTS the \$6.X-debt darwin/arm64 host gap (no /dev/kvm available to podman containers; macOS HVF not exposed to podman) and, when detected, REFUSES to claim a successful gate run. Instead it:

1. Validates arguments + matrix minimum
2. Writes <evidence-dir>/host-preflight.json with the host-gap classification
3. Exits 2 (NOT 0) — gate-host ineligible

Per \$6.J/\$6.L: no false-pass; honest unblock report. Real gate runs require a Linux x86_64 + KVM host.

Usage

```
# Run ALL Challenges on the $6.AE.2 minimum matrix
bash scripts/run-challenge-matrix.sh
```

```
# Run a specific Challenge
bash scripts/run-challenge-matrix.sh \
  --test-class
  lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest
```

```

# Add TV-class AVD (when feature touches TvActivity)
bash scripts/run-challenge-matrix.sh --add-tv

# Add foldable AVD
bash scripts/run-challenge-matrix.sh --add-foldable

# Override "latest stable" API level
bash scripts/run-challenge-matrix.sh --latest-api 36

# Skip APK rebuild
bash scripts/run-challenge-matrix.sh --no-build

# Override the matrix entirely with explicit AVDs (target EXISTING
  host AVDs).
# Use on a macOS host-direct+HVF run where the default CZ_API*
  images are not provisioned.
bash scripts/run-challenge-matrix.sh --avds
  "CZ_API35_Phone_Fresh:35:phone" --no-build

# Single-AVD smoke run against the locally present api34 image
  (the requested
# name is advisory – the runner resolves the image's baked AVD,
  e.g. "default")
bash scripts/run-challenge-matrix.sh --avds
  "CZ_API34_Phone:34:phone" --no-build \
  --test-class lava.app.challenges.Challenge00ColdStartTest

# Pin one explicit image for every AVD (pre-LVA-014 single-image
  behavior)
bash scripts/run-challenge-matrix.sh \
  --container-image ghcr.io/vasic-digital/lava-android-
    emulator:api34-x86_64

# ALSO invoke the 11 HelixQA Challenge scripts (Option 1 wiring)
bash scripts/run-challenge-matrix.sh --include-helixqa

```

Inputs

Arg	Description
--test-class <fqcn>	Specific Challenge to run (default: all under lava.app.challenges)
--evidence-dir <dir>	Output directory (default: dated under .lava-ci-evidence/)
--no-build	Skip the :app:assembleDebug step
--avds "<name:api:form[,...]>"	Replace the §6.AE.2 default matrix entirely with an explicit comma-separated AVD list. Use to target existing host AVDs (e.g. CZ_API35_Phone_Fresh:35:phone on a

Arg	Description
	macOS host-direct+HVF run where the default CZ_API* images are not provisioned). When set, --latest-api / --add-tv / --add-foldable are ignored. A sub-minimum list is a development-iteration run, not a §6.AE.2-conformant gate matrix. NOTE: the Containers CLI still resolves each AVD's system image through tools/lava-containers/vm-images.json (--image-manifest); the manifest must contain a matching image id (e.g. android-35-phone) or the runner fails with no image with id.
--latest-api <N>	Override the "latest stable" API level (default: 36; ignored when --avds is set)
--container-image <ref>	Containerized-runner image. Default: ghcr.io/vasic-digital/lava-android-emulator:api{api}-x86_64 — the {api} token is substituted per AVD api level by the runner (LVA-014 fix #1). A ref without {api} is used verbatim for every AVD.
--container-runtime <podman docker>	Container runtime CLI (default: podman).
--extra-apk <path>	Additional APK installed on each AVD AFTER the client APK, before instrumentation runs. Repeatable — pass more than once to install several companion APKs. Forwarded verbatim, in order, to the Containers CLI's own repeatable --extra-apk flag (MatrixConfig.ExtraAPKPaths). Use for Challenges needing a second app on the SAME device (e.g. Challenge72's on-device api-app companion for same-device mDNS discovery). Added for the §6.X gate wiring that closes the two-app-on-one-emulator capability gap.
--add-tv	Add a TV-class AVD to the matrix
--add-foldable	Add a foldable AVD
--include-helixqa	ALSO invoke scripts/run-helixqa-challenges.sh (the 11 HelixQA Challenge scripts per Option 1 wiring). OFF by default so existing matrix runs are unaffected. HelixQA runs on the HOST BEFORE the AVD matrix →

Arg	Description
	independent of §6.X-debt darwin/arm64 gate-host gap. HelixQA wrapper evidence lands at <evidence-dir>/helixqa/. Non-zero HelixQA exit promotes the final aggregate exit code (matrix exit dominates if both fail). See docs/scripts/run-helixqa-challenges.sh.md for full details.

§6.AE.2 mandatory minimum matrix

CZ_API28_Phone:28:phone
 CZ_API30_Phone:30:phone
 CZ_API34_Phone:34:phone
 CZ_API34_Tablet:34:tablet
 CZ_API<latest>_Phone:<latest>:phone

Plus TV / foldable when explicitly requested by --add-tv / --add-foldable.

Sub-minimums are permitted for development iteration; the gate row's gating: true flag is only set when the full minimum is satisfied + every config dimension (theme/locale/density per §6.AE.2) is covered.

Outputs

- <evidence-dir>/host-preflight.json — host-gap classification (always written)
- <evidence-dir>/real-device-verification.{md,json} — per-AVD attestation rows (only on Linux x86_64 + KVM gate-host)
- <evidence-dir>/helixqa/helixqa-attestation.json + per-script logs — only when --include-helixqa is set

Exit codes

Code	Meaning
0	Matrix completed; all AVDs passed
1	At least one AVD failed boot OR at least one test failed (real Containers CLI exit), OR the LVA-014 image preflight failed (missing image + pull failure — honest error with local-build fallback)
2	Gate-host ineligible (darwin/arm64 OR no KVM); honest unblock-needed report written

Cross-references

- `submodules/containers/cmd/emulator-matrix` (the underlying runner)
- `tools/lava-containers/vm-images.json` (matrix manifest)
- `scripts/run-emulator-tests.sh` (older sister-glue; same delegation, different default arguments)
- `scripts/run-helixqa-challenges.sh` + `docs/scripts/run-helixqa-challenges.sh.md` (the HelixQA wrapper invoked by `--include-helixqa`)
- `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 wiring design)
- `.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json` (the standing §6.X-debt incident)
- `Lava CLAUDE.md` §6.AE + §6.X + §6.I